University of Ottawa

Department of Mathematics and Statistics

MAT 2143: Algebraic Structures

Professor: Hadi Salmasian

Final Exam

April 2016

Surname _____ First Name _____

Student # _____

Instructions:

(a) You have 3 hours to complete this exam.

(b) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.

(c) Write your student number at the top of each page in the space provided.

(d) No notes, books, scrap paper, calculators or other electronic devices are allowed.

(e) You are strongly recommended to write in **pen**, not pencil.

(f) You may use the last page of the exam as scrap paper.

(g) Cellular phones, unauthorized electronic devices or course notes (unless an open-book exam) are not allowed during this exam. **Phones and devices must be turned off and put away in your bag**. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: you will be asked to leave immediately the exam and academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.

By signing below, you acknowledge that you have ensured that you are complying with the above statement.

Signature _____

Please do not write in the table below.

Question	1	2	3	4	5	6	7	8	9	10	11	Total
Maximum	5	3	4	5	5	5	6	3	3	5	4	48
Grade												

1. **[3 points]** Determine all $x \in \mathbb{Z}$ such that $x \equiv 3 \pmod 7$ and $x \equiv 2 \pmod 8$.

Solution: We need to find $p, q \in \mathbb{Z}$ such that $7p + 8q = 1$. For example $p = -1$ and $q = 1$. Then

$$x \equiv (3)(8)q + (2)(7)p = 24 - 14 = 10 \pmod{56}.$$

(b) **[2 points]** Using Fermat's Theorem, calculate the residue of 2^{161} in division by 17.

Solution: We have $2^{16} \equiv 1 \pmod{17}$ and therefore since $161 = 16 \times 10 + 1$, we obtain

$$2^{161} \equiv \left(2^{16}\right)^{10} (2^1) \equiv 2 \pmod{17}.$$

2. **[3 points]** Let p be an odd prime number, and assume that $n \in \mathbb{Z}$ is such that $p^2|n^3-n$. Prove that either $p^2|n$ or $p^2|n-1$ or $p^2|n+1$.

3. (a) **[2 points]** Write down the decomposition of the following permutation in S_9 as a product of disjoint cycles:

$$(1\ 3\ 7\ 6)(2\ 3\ 4)^{-1}$$

Solution: $(1\ 3\ 2\ 4\ 7\ 6)$

In the following two parts, the commutator $[\sigma, \tau]$ of two elements $\sigma, \tau \in S_n$ is defined by

$$[\sigma, \tau] = \sigma\tau\sigma^{-1}\tau^{-1}.$$

(b) **[1 point]** Give an example of two permutations $\sigma, \tau \in S_3$ such that $[\sigma, \tau] \neq \varepsilon$. (As usual, ε denotes the identity element of S_n.) You should justify that your example is valid.

Solution: For example $(1,2)(1,3)(1,2)^{-1}(1,3)^{-1} = (1\ 2\ 3) \neq \varepsilon$.

(c) **[1 point]** Determine whether or not it is possible to find some $n \geq 2$ and permutations $\sigma, \tau \in S_n$, such that $[\sigma, \tau] = (1\ 2)$. Justify your answer.

4. (a) **[2 points]** Write down the definition of a normal subgroup of a group G.

(b) **[3 points]** Let G be a group and let N be a normal subgroup of G. Let H be another group. Prove that $N \times H$ is a normal subgroup of $G \times H$.

Solution: Being a subgroup follows from Subgroup test. Being normal is straightforward.

5. (a) **[2 points]** Let G be a group. Write down the definition of the order of an element $g \in G$.

(b) **[3 points]** Write down the proof of the following theorem:

If G is a finite group and $g \in G$, then $o(g)$ divides $|G|$.

6. (a) **[3 points]** Write down all subgroups of $\mathbb{Z}_{18}$ explicitly. You should justify your answer.

(b) **[2 points]** Let $g \in C_{22}$ be a generator for the cyclic group C_{22} (recall that $C_{22} \simeq \mathbb{Z}_{22}$), and set $H := \langle g^4 \rangle$. Determine the index $|C_{22} : H|$. You should justify your answer.

Solution: $\mathbb{Z}_{22}$ is generated by $g^{\gcd(4,22)}$, i.e., by g^2. Thus $|\mathbb{Z}_{22} : H| = 2$.

7. Let $\phi : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ be the map defined by

$$\phi(x, y) := 2x - 4y \text{ for every } (x, y) \in \mathbb{Z} \times \mathbb{Z}.$$

(a) **[2 points]** Prove that ϕ is a group homomorphism.

(b) **[2 points]** Determine $\ker(\phi)$ explicitly.

(c) **[2 points]** Using the Isomorphism Theorem, prove that $(\mathbb{Z} \times \mathbb{Z})/\ker(\phi) \cong \mathbb{Z}$.

8. (a) **[1 point]** Write down the definition of the centre of a group G.

(b) **[2 points]** Prove the following statement:
If $\alpha : G \to H$ is a group homomorphism, then $\ker(\alpha)$ is a normal subgroup of G.

9. **[3 points]** Let G be a group and let $K \subseteq G$ be a normal subgroup of G such that $|G : K| = m$, where $m \in \mathbb{N}$. Prove that for every $g \in G$, we have $g^m \in K$.

10. (a) **[2 points]** Give an example of a non-commutative ring. You need to justify that your example is valid.

(b) **[2 points]** Write down the definition of the characteristic of a ring.

(c) **[1 point]** Determine the characteristic of the ring $\mathbb{Z} \times \mathbb{Z}_4$.

11. [**4 points**] For each of the following statements, write "**T**" if the statement is true, and write "**F**" if it is false. You do not need to justify your answer. For each correct you will receive .5 points, and for each incorrect answer you will receive $-.25$ points.

____ For a subgroup H of a group G, it is always true that for every $a, b \in G$, if $Ha = Hb$ then $aH = bH$.

Solution: F

____ Every subgroup of an abelian group G is normal in G.

Solution: T

____ The group S_n is simple for every $n \geq 3$.

Solution: F

____ If H and K are two normal subgroups of a group G, then $H \cap K$ is also a normal subgroup of G.

Solution: T

____ For every $n \geq 2$, the dihedral group D_n has a cyclic subgroup of order n.

Solution: T

____ For a group G and subgroups $H, K \subseteq G$, if H is normal in G then $HK = KH$.

Solution: T

____ There exists a field of characteristic 6.

Solution: F

____ Every cyclic group is isomorphic to $\mathbb{Z}_n$ for some $n \in \mathbb{N}$.

Solution: F